Supplemental Material

Appendix A
Taxonomy of LDoS Attacks

LDoS attacks can be categorized by their exploitation mechanisms and targeted protocol layers. We group representative attacks into three categories: (i) HTTP connection exhaustion, (ii) transport-level manipulation, and (iii) protocol exploitation attacks.

HTTP Connection Exhaustion Attacks:

Slowloris maintains numerous incomplete HTTP connections by sending partial headers without terminating characters `\r\n\r\n`, keeping connections open and exhausting server resources [1], [2]. Similarly, **Slow POST** sends HTTP POST requests with large `Content-Length` headers but transmits minimal data, forcing the server to hold connections [3]. **SlowReq** extends connection lifetimes by gradually sending incomplete HTTP requests, exploiting server timeout mechanisms [4]. **Slow NEXT** abuses persistent HTTP connections by introducing long delays between valid requests, exhausting server concurrency limits [5].

Transport-Level Manipulation Attacks:

Slow Read requests large resources while advertising an abnormally small TCP receive window, forcing slow data transmission and prolonged sessions [6]. **SlowDrop** intermittently drops server responses to trigger retransmissions, mimicking poor network conditions [7]. **SlowComm** transmits fragmented or malformed packets, leaving the server waiting for additional data and consuming buffer resources [8].

Protocol Exploitation Attacks:

Apache Range Header exploits Apache's handling of overlapping byte-range requests by issuing a single request with many overlapping ranges, significantly increasing server workload [9].

Appendix B
Feature Selection

We identify a compact set of high-predictive features for efficient and generalizable LDoS detection using feature importance ranking, forward elimination, and cross-validation. An initial feature set is extracted, and importance scores are computed using RF model (TABLE I). We then apply forward elimination, iteratively removing lower-ranked features while evaluating performance via cross-validation. Fig. 2 shows that detection accuracy saturates when using the top-ranked features, indicating diminishing returns from additional features. The final feature set (F1-F5) represents the most informative and stable subset. Lower-ranked features (F6-F7) provide auxiliary signals in specific scenarios, such as distinguishing traffic patterns under constrained observability (e.g., encrypted, low-variance flows), while F8 captures coarse inactivity characteristics with limited discriminative value. However, their overall contribution is marginal relative to their computational cost in streaming environments. These features are therefore excluded to maintain a compact and efficient design without degrading detection performance.

TABLE I: Feature Importance Scores

ID	Feature Name	Feature Breakdown	Score
F1	Inter-arrival Time	Inter-arrival Time (Max, Min, Std)	867
F2	Window Size	Window Size (Fwd_Byts, Bwd_Byts)	745
F3	Packet Length	Packet Length (Max, Min, Mean, Std, Fwd, Bwd)	728
F4	Connection Density Per IP	—	630
F5	Flow Duration	—	549
F6	Packets Count	Packets Count (Fwd Pkts/s, Bwd Pkts/s)	451
F7	Up/Down Ratio	—	319
F8	Idle Max	—	236
:	:	:	:

Appendix C
Simulation of LDoS Attacks and Server Response Analysis

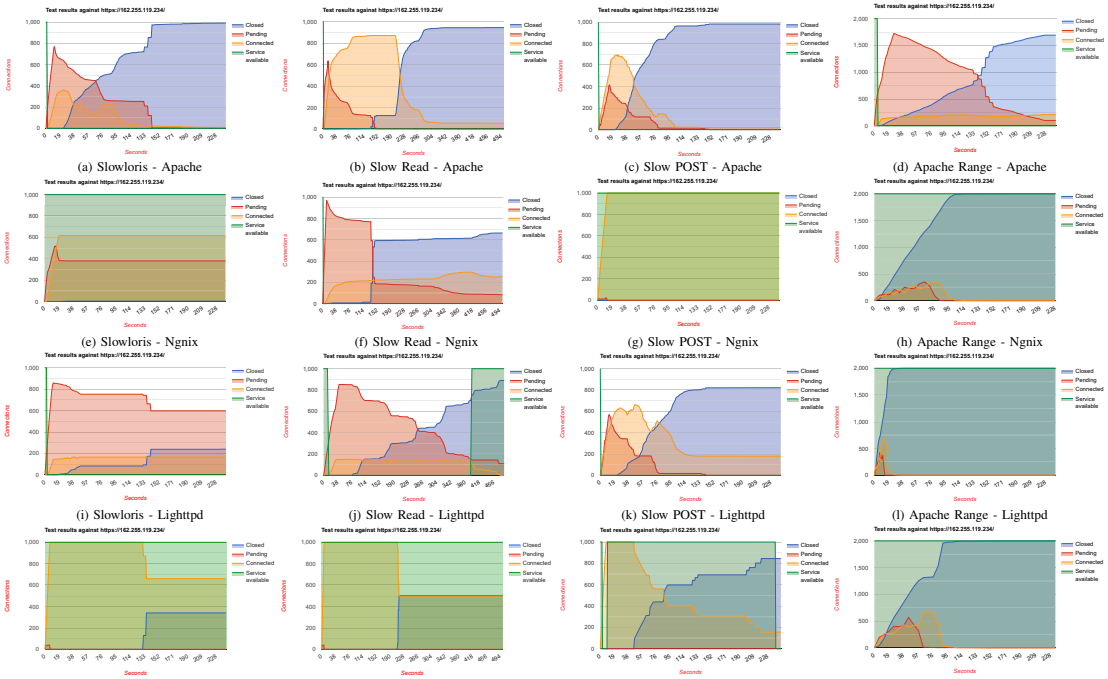
We summarize the results of simulated LDoS attacks conducted to evaluate the resilience of commonly deployed web servers under controlled experimental settings [9]. The experiments (Fig. 1a-1p) were performed in a realistic testbed with varying attack parameters to emulate different operating conditions. On Apache (Fig. 1a), the Slowloris attack initiated 1,000 connections at 200 connections/s over 240 s, causing service degradation within $\approx$ 5 s. Similarly, Slow POST, Slow Read, Slow NEXT, SlowDrop, SlowComm, SlowReq, and Apache Range attacks caused service unavailability early in their progression, preventing legitimate access throughout the test duration. In contrast, Nginx showed higher resilience, managing and recovering under most attacks without affecting service availability. As shown in Figs. 1e, 1f, 1g, and 1h, it remained stable under Slowloris, Slow POST, SlowDrop, SlowReq, and Apache Range attacks, likely due to its event-driven architecture. However, Slow Read, Slow NEXT, and SlowComm still caused service disruption. Lighttpd exhibited higher susceptibility across most attack types. As shown in Figs. 1i, 1j, and 1k, attacks such as Slowloris, Slow POST, Slow Read, Slow NEXT, SlowDrop, SlowComm, and SlowReq led to service disruption, while Apache Range (Fig. 1l) failed to disrupt service availability. MS IIS showed the highest robustness among the evaluated servers. Under IIS 10.0, none of the tested attacks caused sustained service disruption despite persistent attack attempts (Figs. 1m, 1n, 1o, and 1p). However, SlowDrop remained effective against IIS 8.0.

TABLE II summarizes the results across servers and their versions. A successful attack ("✓") indicates service disruption lasting over 10 s whereas an unsuccessful attack ("✗") indicates no sustained disruption. The results show that widely deployed web servers remain vulnerable to specific LDoS strategies, highlighting the need for adaptive detection mechanisms against stealthy resource-exhaustion attacks.

TABLE II: Attack Simulation Results

Attack	Apache (2.4.52)	Apache (2.4.53)	Apache (2.4.58)	Nginx (1.14.2)	Nginx (1.18.0)	Lighttpd (1.4.59)	Lighttpd (1.4.64)	IIS (8.0)	IIS (10.0)
Slowloris	✓	✓	✓	✗	✗	✓	✓	✗	✗
Slow POST	✓	✓	✓	✗	✗	✓	✓	✗	✗
Slow Read	✓	✓	✓	✓	✓	✓	✓	✗	✗
Slow NEXT	✓	✓	✓	✓	✓	✓	✓	✗	✗
SlowDrop	✓	✓	✓	✗	✗	✓	✓	✓	✗
SlowComm	✓	✓	✓	✓	✓	✓	✓	✗	✗
SlowReq	✓	✓	✓	✗	✗	✓	✓	✗	✗
Apache Range	✓	✓	✓	✗	✗	✗	✗	✗	✗

✓DoS Obtained, ✗ DoS Not Obtained



(a) Slowloris - Apache (b) Slow Read - Apache (c) Slow POST - Apache (d) Apache Range - Apache

(e) Slowloris - Ngnix (f) Slow Read - Ngnix (g) Slow POST - Ngnix (h) Apache Range - Ngnix

(i) Slowloris - Lighttpd (j) Slow Read - Lighttpd (k) Slow POST - Lighttpd (l) Apache Range - Lighttpd

(m) Slowloris - MS IIS (n) Slow Read - MS IIS (o) Slow POST - MS IIS (p) Apache Range - MS IIS

Fig. 1: Experimental Results of Attack Simulations

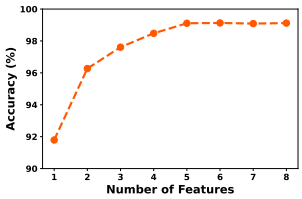


Fig. 2: Impact of Features on Accuracy

Appendix D
Evaluation Metrics

TABLE III shows formulas for commonly used metrics. In these formulas, True Positives (TP) denote attack instances correctly detected, True Negatives (TN) denote benign instances correctly classified, False Positives (FP) denote benign instances misclassified as attacks, and False Negatives (FN) denote attack instances that are missed. FPR and FNR quantify false alarms and missed detections, respectively. Precision measures the correctness of predicted attack instances, while recall measures the proportion of actual attacks that are detected. F1-Score provides a harmonic balance between precision and recall, and accuracy reflects the overall classification performance. For multi-class classification, these metrics are computed using a one-vs-rest scheme with macro-averaging to ensure equal contribution from each class.

TABLE III: Evaluation Metrics

Metric	Formula
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$
False Negative Rate (FNR)	$\frac{FN}{FN+TP}$
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1-Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}}$

Appendix E
Evaluation of ITC and ATC Models

The experimental results of both the ITC and ATC models are summarized in Table IV and Table V.

TABLE IV: Performance Evaluation of ITC Models

w	DT					
	Prec (%)	Rec (%)	F1 (%)	Acc (%)	FPR (%)	FNR (%)
2^{18}	99.24	99.32	99.28	99.31	0.054	0.58
w	RF					
	Prec (%)	Rec (%)	F1 (%)	Acc (%)	FPR (%)	FNR (%)
2^{18}	99.35	99.48	99.40	99.42	0.047	0.45
w	SVM					
	Prec (%)	Rec (%)	F1 (%)	Acc (%)	FPR (%)	FNR (%)
2^{18}	**99.72**	99.51	99.54	99.56	**0.018**	0.49
w	LSTM					
	Prec (%)	Rec (%)	F1 (%)	Acc (%)	FPR (%)	FNR (%)
2^{18}	99.50	**99.68**	**99.63**	**99.67**	0.029	**0.24**

Prec: Precision, **Rec**: Recall, **F1**: F1-Score, **Acc**: Accuracy

References

[1] V. Rios, P. Inacio, D. Magoni, and M. Freire, "Detection of slowloris attacks using machine learning algorithms," in *Proceedings of the 39th ACM/SIGAPP symposium on applied computing*, 2024, pp. 1321–1330.

[2] E. Cambiaso, G. Papaleo, G. Chiola, and M. Aiello, "Slow dos attacks: definition and categorisation," *International Journal of Trust Management in Computing and Communications*, vol. 1, no. 3-4, pp. 300–319, 2013.

[3] Cloudflare, "R u dead yet? (r.u.d.y.) attack," 2024. [Online]. Available: https://www.cloudflare.com/learning/ddos/ddos-attack-tools/r-u-dead-yet-rudy/

[4] M. Sikora, R. Fujdiak, K. Kuchar, E. Holasova, and J. Misurec, "Generator of slow denial-of-service cyber attacks," *Sensors*, vol. 21, no. 16, p. 5473, 2021.

[5] E. Cambiaso, G. Papaleo, G. Chiola, and M. Aiello, "Designing and modeling the slow next dos attack," in *International Joint Conference: CISIS'15 and ICEUTE'15*. Springer, 2015, pp. 249–259.

[6] C. Kemp, C. Calvert, and T. M. Khoshgoftaar, "Detection methods of slow read dos using full packet capture data," in *2020 IEEE 21st International Conference on Information Reuse and Integration for Data Science (IRI)*. IEEE, 2020, pp. 9–16.

[7] E. Cambiaso, G. Chiola, and M. Aiello, "Introducing the slowdrop attack," *Computer Networks*, vol. 150, pp. 234–249, 2019.

[8] E. Cambiaso, G. Papaleo, and M. Aiello, "Slowcomm: Design, development and performance evaluation of a new slow dos attack," *Journal of Information Security and Applications*, vol. 35, pp. 23–31, 2017.

TABLE V: Performance Evaluation of ATC Models

w	Model	Attack Type	Prec(%)	Rec(%)	F1(%)	Acc(%)	FPR(%)	FNR(%)
2^{18}	DT	Slowloris	98.16	98.89	98.53	-	0.21	1.11
		Slow POST	99.94	99.89	99.54	-	0.00	0.01
		Slow Read	97.66	97.80	97.88	-	0.24	2.20
		SlowComm	97.82	98.63	98.21	-	0.32	1.26
		Slow NEXT	99.39	99.74	99.52	-	0.14	0.16
		SlowDrop	98.40	98.56	98.48	-	0.18	1.44
		SlowReq	99.23	98.03	98.67	-	0.20	1.98
		Apache Range	99.08	99.21	99.19	-	0.15	0.85
		Overall	98.91	98.86	98.85	98.90	0.26	1.19
2^{18}	RF	Slowloris	99.42	99.63	99.45	-	0.12	0.26
		Slow POST	99.49	99.36	99.41	-	0.11	0.41
		Slow Read	98.44	98.68	98.65	-	0.19	1.30
		SlowComm	99.12	98.21	98.69	-	0.12	1.77
		Slow NEXT	99.05	98.86	99.01	-	0.25	1.06
		SlowDrop	97.48	98.20	98.32	-	0.33	0.82
		SlowReq	98.80	99.24	99.08	-	0.18	0.67
		Apache Range	99.38	99.40	99.35	-	0.10	0.59
		Overall	99.06	99.10	99.08	99.10	0.18	1.02
2^{18}	SVM	Slowloris	99.15	98.33	99.12	-	0.09	1.67
		Slow POST	99.68	99.71	99.70	-	0.02	0.10
		Slow Read	98.26	97.94	98.59	-	0.21	2.06
		SlowComm	97.83	98.90	98.35	-	0.32	1.10
		Slow NEXT	99.53	99.56	99.54	-	0.07	0.20
		SlowDrop	98.41	99.04	98.72	-	0.17	0.96
		SlowReq	98.95	99.08	99.02	-	0.15	0.94
		Apache Range	99.84	99.85	99.68	-	0.01	0.00
		Overall	99.15	99.12	99.16	99.14	0.15	0.92
2^{18}	LSTM	Slowloris	99.54	99.26	99.53	-	0.05	0.68
		Slow POST	99.84	99.90	99.85	-	0.02	0.08
		Slow Read	98.80	98.27	99.08	-	0.22	1.78
		SlowComm	98.68	99.49	99.19	-	0.18	0.41
		Slow NEXT	99.71	99.73	99.57	-	0.03	0.10
		SlowDrop	99.38	99.26	99.42	-	0.21	0.86
		SlowReq	99.23	98.82	99.26	-	0.13	1.18
		Apache Range	99.85	99.88	99.72	-	0.02	0.09
		Overall	**99.34**	**99.26**	**99.25**	**99.28**	**0.10**	**0.67**

Prec: Precision, **Rec**: Recall, **F1**: F1-Score, **Acc**: Accuracy

[9] H. Tang, "Evaluation of http ddos cyber attack on web servers apache and nginx," 2024. [Online]. Available: https://hdl.handle.net/10652/6499